LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Daily Practice: Ch 9 Light – Reflection and Refraction

Chapter / Topic: Light – Reflection and Refraction · Week 5 of the 90-day plan

Name: _____ Roll No.: _____ Date: _____

Marks obtained: _____

General instructions:

1. One day at a time, about 25 minutes each: Day 1 on the day you study the chapter, then Day 2 and Day 3. Write answers in your notebook, check them with the key the same day, and copy every mistake into your error log.
2. Numericals: write u, v, f with signs first (u is negative). Mirror: 1/v + 1/u = 1/f, m = −v/u. Lens: 1/v − 1/u = 1/f, m = v/u.

DAY 1: SPHERICAL MIRRORS

Q1. The focal length of a concave mirror is 15 cm. Its radius of curvature is:

(a) 7.5 cm (b) 15 cm
(c) 30 cm (d) 45 cm

Q2. An object is placed at the centre of curvature (C) of a concave mirror. The image is:

(a) at F; real, diminished
(b) beyond C; real, enlarged
(c) behind mirror; virtual, erect
(d) at C; real, inverted, same size

Q3. The mirror formula is:

(a) 1/v − 1/u = 1/f (b) m = v/u
(c) u + v = f (d) 1/v + 1/u = 1/f

Q4. A ray of light passing through the centre of curvature of a concave mirror, after reflection:

(a) retraces its path (b) passes through the focus
(c) becomes parallel to the axis (d) passes through the pole

Q5. State the New Cartesian sign convention for spherical mirrors.

Q6. An object is placed 30 cm in front of a concave mirror of focal length 20 cm. Find the position of the image, the magnification and the nature of the image.

Q7. A car's convex rear-view mirror has a radius of curvature of 4 m. A motorbike is 6 m behind the car. Find the position, nature and magnification of the image of the motorbike.

Q8. An object 3 cm high is placed 6 cm in front of a concave mirror of focal length 12 cm. Find the position, size and nature of the image.

DAY 2: REFRACTION AND LENSES

Q9. The refractive index of water is 1.33. This means that:

(a) light travels 1.33 times faster in water than in vacuum
(b) the speed of light in vacuum is 1.33 times its speed in water
(c) water is 1.33 times denser than air
(d) light bends by 1.33° on entering water

Q10. A lens has a power of −2.5 D. It is a:

(a) convex lens, f = 40 cm (b) concave lens, f = 40 cm
(c) concave lens, f = 2.5 m (d) convex lens, f = 25 cm

Q11. A convex lens forms a real, inverted image of the same size as the object when the object is placed:

(a) at F_1 (b) between F_1 and $2F_1$
(c) at $2F_1$ (d) at infinity

Q12. A ray of light passing through the optical centre of a lens:

(a) bends towards the axis (b) becomes parallel to the axis
(c) goes through undeviated (d) is reflected back

Q13. Why does a ray of light bend when it passes obliquely from air into glass? What happens to a ray that falls along the normal?

Q14. The speed of light in a liquid is 2.25×10^8 m/s. Find its refractive index. (Speed of light in vacuum = 3×10^8 m/s.) Which common liquid could it be?

Q15. An object is placed 12 cm from a convex lens of focal length 8 cm. Find the position of the image, the magnification and the nature of the image.

Q16. An object is placed 15 cm from a concave lens of focal length 10 cm. Find the position of the image, the magnification and the nature of the image.

DAY 3: MIXED PRACTICE AND CHALLENGE

Q17. **Assertion (A):** A concave mirror can form both real and virtual images.
Reason (R): The nature of the image formed by a concave mirror depends on where the object is placed.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

Q18. Which of these always forms a virtual, erect and diminished image, wherever the object is?

(a) a concave mirror (b) a convex mirror
(c) a convex lens (d) a plane mirror

Q19. A lens gives a magnification of −1. The image is:

(a) virtual, same size (b) real, enlarged
(c) virtual, diminished (d) real, inverted, same size

Q20. A student focused the image of a distant building on a screen using a convex lens. The screen was 20 cm from the lens. She then placed a candle 30 cm in front of the same lens.

(i) What is the focal length of the lens? How do you know?

(ii) Where must she place the screen to get a sharp image of the candle?

(iii) Find the magnification and describe the image.

Q21. (a) State the paths of three rays you can use to draw ray diagrams for a convex lens.
(b) A 5 cm tall object is 15 cm in front of a convex mirror of focal length 10 cm. Find the position, height and nature of the image.

Q22. A concave mirror and a convex mirror look the same from the back. How can you tell them apart just by looking into them?

~ Sign convention right = numerical right. Do one extra numerical tonight! ~